

The Complete Nature of the Collatz Obstruction

From the Pointwise Valuation Problem to the Shadow Frontier:
what was eliminated, what remains, and what new mathematics is required

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Abstract

This document is the capstone synthesis of a multi-year Collatz program spanning three campaigns: the pointwise-valuation/SRCE program (the `collatz/` manuscripts), the finite-spectral-shadow program (`deterministic_equidistribution_theory/`), and the Shadow Frontier (`shadow_frontier/`, the endpoint). No proof of the conjecture is claimed. The claim is sharper and, we believe, more useful: *the obstruction has been located, compressed, priced, and shown to be invariant under change of framework*. Three independent routes — ours, Tao’s almost-all theory, and a contemporaneous LLM-collaboration program audited herein — terminate, under explicit dictionaries, at the same residual statement. In its most compressed form the obstruction is one scalar parity sequence: the survival readings $v_2(3^\ell m - 1) = v_2(m - 3^{-\ell})$ of the two-clock round chain, provably i.i.d. fair-thirds on Haar fibers, provably deterministic (and lethal) on ghost-aligned fibers, with every structured deviation metered by exact clocks, without-replacement supply, entropy deficits, and an amortized census budget. What is missing is exactly one species of theorem: a *distributional-to-pointwise* (quenched) rigidity statement for a single self-referential zero-entropy stream. We trace how each campaign eliminated its layer of candidate mechanisms, state the endpoint constraints with their constants ($D^* = 1.2463$, $\varepsilon_{\text{bin}} = 0.449$, $\varepsilon_{\text{nat}} = 0.1397$, $R = 0.0882$, $\theta_{12} \leq 0.97907$), and close with the new mathematics required, keyed to canonical and contemporary references.

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1 Accelerated Syracuse notation

We work with the odd accelerated Syracuse map

$$T(x) = \frac{3x+1}{2^{b(x)}}, \quad b(x) = v_2(3x+1),$$

x odd and positive unless stated otherwise. Along an orbit write $b_n = b(x_n)$, $B_N = \sum_{n < N} b_n$. The critical slope is $\log_2 3$, and the logarithmic identity

$$\log_2 x_N = \log_2 x_0 + N \log_2 3 - B_N + \sum_{n < N} \log_2 \left(1 + \frac{1}{3x_n}\right) \quad (1)$$

is exact. Mean letter excess $\frac{1}{N}B_N > \log_2 3 + \eta$ forces descent (the C1 mechanism). For a finite character χ' the two-point shadow satisfies the exact keystone identity

$$F_{\chi'}(x_n, x_{n+1}) = \chi'(x_{n+1}) \overline{\chi'(3x_n+1)} = \chi'(2)^{-b_n}, \quad (2)$$

with geometric stationary value $G(z) = \sum_{b \geq 1} 2^{-b} z^{-b}$ for $z = \chi'(2)$. Under Haar measure on $\mathbb{Z}_2$ the letters are i.i.d. geometric, $\Pr[b = k] = 2^{-k}$, $\mathbb{E}[b] = 2 > \log_2 3$: a typical orbit descends. The entire difficulty of the conjecture is the upgrade of this annealed statement to every single deterministic orbit.

2 Campaign I: the pointwise valuation problem and SRCE

The first program (`collatz/docs/`: the master dependency document, the pointwise-valuation paper, the stabilized-residue extinction paper, and the sparse-target pivot note) isolated the quenched upgrade as a standalone statement and then localized it.

2.1 The conditional reduction and the annealed theorem

The honest logical skeleton, recorded in the master dependency document, is one-directional:

$$\text{SRCE} + \text{Exit Inequality} + \text{cycle elimination} \implies \text{Collatz}.$$

The *weak Pointwise Valuation Theorem* (PVT) — the sole pointwise input — was proved in its *annealed* form with an explicit machine-checked Lyapunov function (a correction-weighted, floor-killed transfer operator of spectral radius $\leq \sqrt{3}/(2\sqrt{2}-1) < 1$). The canonical-residue formula

$$r_m \equiv 3^{-m}(2^{D_m} - A_m) \pmod{2^{D_m+1}}$$

showed the weak PVT *equivalent* to a deterministic shrinking-target statement, the *Stabilized Residue Cluster Extinction* problem (SRCE): no fixed odd x_0 admits an infinite valuation word whose canonical residues stabilize to x_0 while the deficit $D_m - m \log_2 3$ stays confined.

2.2 Why Borel–Cantelli fails, and the H23a localization

The targets have summable density $\asymp 3^{-m}$ and the residue map is per-scale uniform — yet ordinary Borel–Cantelli is powerless, for an identified structural reason: the lift digits are nonnegative, the residue sequence is *monotone*, and hence the deterministic target hits are *maximally clustered*. The extremal set was represented as a peak-rooted inverse-Syracuse tree, modeled as a killed critical branching random walk in a strip, and SRCE was reduced to a fixed-modulus Perron–Frobenius gap for the exact deterministic congruence operator (H23a), with the annealed gap a theorem and the character-killing mechanisms partially machine-checked in Lean.

2.3 The lesson of Campaign I

Every natural mechanism was eliminated by proof or computation: finite-state and block Lyapunov functions, Baker/convergent near-cycle control, word-level entropy and complexity, $\times 2 \times 3$ entropy rigidity, p -adic scenery overlap, restart-density arguments. The postmortem (`why_collatz_fails.md`) identified the wall exactly: the confined counterexample word is *symbolically generic* — indistinguishable from a confined renewal word in every word statistic — so no annealed or finite-state mechanism can see it. The sparse-target pivot note drew the correct conclusion: the remaining obstruction is a *deterministic sparse-target avoidance problem for a non-invertible affine cocycle*, demanding an inverse-theorem framework (persistent over-hitting must force finite algebraic/coboundary structure) rather than another estimate. This diagnosis fixed the agenda for everything that followed.

3 Campaign II: finite spectral shadows

The second program built the inverse-theorem infrastructure on the finite (annealed) side and proved the global reduction arrow.

3.1 Coboundary classification and the finite gap

The finite coboundary classification $\text{Cob} = \langle \chi_2 \rangle$ (parity is the only coboundary direction; the proof uses the $b \mapsto b+2$ branch comparison and the primitivity of 2 modulo powers of 3) separates genuine shadows from the parity escape. On that foundation:

Theorem 1 (Finite non-coboundary spectral gap, informal). *At every finite conductor level, every primitive non-coboundary twisted synchronized transfer operator has spectral radius strictly below the untwisted operator, and the high-conductor sector has a uniform gap at $s = 0$ (numerically $\sup_c \gamma_c(0) \leq (\sqrt{3}/2)^{1/2} + o(1) \approx 0.9306$).*

The proof required the coboundary classification, conductor collapse, exact pair-correlation phase identities, a threshold cascade, bounded-shell flattening, and a root-ball/Hensel analysis of the degenerate cells. It is an *annealed* result: the operator mixes shadows; no single orbit is yet implicated.

3.2 The keystone arrow and quenched shadow exclusion

Theorem 2 (Keystone arrow, informal). *A bad positive orbit (divergent, or trapped in a nontrivial cycle) must, for some finite non-coboundary χ' , have empirical two-point shadow averages failing to converge to $G(\chi'(2))$.*

By (2) the shadows *are* letter characters, so the statement is finite. The arrow is one-directional (the trivial cycle itself deviates), and the corrected global endpoint is *Quenched Shadow Exclusion*: no non-eventually-trivial positive orbit carries a persistent non-coboundary shadow — which is Collatz in finite-shadow language.

3.3 Repetition rigidity and the exclusions

The genuinely new theorem of Campaign II, machine-verified in Lean 4:

Theorem 3 (2-adic repetition rigidity). *Two positions of a positive orbit carrying the same length- p valuation block, lying within 2^{p+1} of each other, are equal; the orbit is then an integer cycle.*

Combined with the Baker/Rhin two-logarithm bound ($x_{\min} \leq Cp^{\tau+1}$, $\tau = 13.3$) this kills all cycles in verified ranges and all low-complexity aperiodic ghosts: Sturmian, Beatty, morphic, linearly recurrent, polynomial-complexity, and bounded-discrepancy valuation words are not realizable by divergent positive orbits. Mean-visible deviations die by (1). The 3-adic-only rigidity hope is *false* (the all-ones trap is coherent, biased, certificate-free modulo every 3^r , and is the ghost $x = -1$); the 2-adic realization tower carries the positivity obstruction and is not auxiliary.

4 Campaign III: the Shadow Frontier (the endpoint)

The third campaign (`shadow_frontier/`) converted the surviving-profile description into exact, certified, machine-verified structure. Every constant below is computed by exact integer arithmetic or verified at scale.

4.1 Exact counting and the phase diagram

D -confinement quantizes the partial sums B_j into integer windows of size $\leq \lfloor 2D \rfloor + 1$, so confined-word counts $N_D(p)$ are *exactly* computable (no Perron estimates). The narrow-band dichotomy follows: an acyclic orbit dwells at most $CY^{\theta(D)}$ steps in a band $[Y, 2^{2D}Y]$, with counting capacity exhausted at

$$D^* = 1.2463 \quad (\text{band ratio } 2^{2D^*} = 5.63).$$

At ratio ≤ 2 the dwell bound is *logarithmic* (the dyadic strip carries exactly one word per phase). Tilting by the shadow observable yields the exact (D, ε) phase diagram, whose structure is now mapped: the vertical counting wall at D^* ; the horizontal *binary bias ceiling* $\varepsilon_{\text{bin}} = |(2 - \log_2 3)\zeta_3^{-1} + (\log_2 3 - 1)\zeta_3 - G| = 0.44903$; the *free-bias basin* below $\varepsilon_{\text{nat}}(D) \rightarrow |\mathbb{E}_{\text{cal}} - G| = 0.1397$ (bias below the basin floor costs nothing; above it the Legendre tilt t^* is the exact price, $\partial\theta/\partial\varepsilon = -t^*/\ln 2$); genuine arithmetic kinks at $2D \in \mathbb{Z}$ and Diophantine resonance lines $D = |m - j \log_2 3|$; and a

first-order *direction-exchange corner* at $(D, \varepsilon) = (2.11, 0.402)$ where the optimal conspiracy direction jumps from the binary $\{1, 2\}$ channel to the phase-coherent $\{1, 4\}$ channel. A counterexample's banded windows are bracketed to $[1.2463, \approx 4.08] \times (0, \varepsilon^*(D)]$, $\varepsilon^* \leq 0.54$.

4.2 The two clocks and the universal ghost clock

Every letter is a 2-adic depth reading: $b = 1$ iff $x \equiv 3 \pmod{4}$, and a maximal 1-run has length *exactly* $v_2(x + 1) - 1$ (the -1 clock); $b = 2$ iff $x \equiv 1 \pmod{8}$, with run length $\lfloor (v_2(x - 1) - 1)/2 \rfloor$, depth decrement exactly 2, and a parity handoff (the $+1$ clock — the trivial cycle is the $+1$ ghost); $b \geq 3$ iff $x \equiv 5 \pmod{8}$, reading $b = v_2(x + \frac{1}{3})$. This generalizes to *every* signature σ of depth K : the phantom root $\rho_\sigma = C_\sigma / (2^K - 3^\ell)$ (the catalogued ghosts $-1, +1, -5, -19/11, -17$ are instances) carries an exact clock — alignment drops exactly K per ridden block, rides last exactly $\lfloor (a - 1)/K \rfloor$ blocks, and alignment a at height x costs $2^a \leq |2^K - 3^\ell|x + C_\sigma$. Consequences, all unconditional: without-replacement depth supply $(2^{k-g}$ pool points per band per ghost, consumed once by an acyclic orbit); at most 2^{k-3} upward crossings of level 2^k , ever; a divergent orbit spends $O(X \log X)$ lifetime steps below X . Amortized over *all* ghost families of depth ≥ 3 , the available drift gain is $R = 0.0882$ bits per letter against the 0.415-bit budget ($4.7\times$ margin) — independently recomputed in the audit of the contemporaneous program of Chang [36].

4.3 The round chain, the i.i.d. coin, and the LTE cascade

Parsed at 1-run starts, survival against the letters- ≤ 2 regime is the reading

$$v = v_2(3^\ell m - 1) = v_2(m - 3^{-\ell}),$$

death iff v odd: an alignment reading of fresh seed bits against a target rotating through $\langle 3 \rangle \subset \mathbb{Z}_2^\times$. On Haar fibers the readings are *exactly* i.i.d. (survival $\frac{1}{3}$ per round; the proof composes the orbit-parameterization slope $2 \cdot 3^B$ with an exact census lemma); apparent finite-modulus anti-correlations are truncation artifacts. Surviving rounds drift *upward* at $+0.6166$ bits per round: the 2-adic Cantor set of survivors is measure zero and its positive members, if any, are forced ascenders. On ghost-aligned fibers the coin is *deterministic*: lifting-the-exponent decides Mersenne seeds ($2^g - 1$ survives round one iff g is odd with $v_2(g - 1)$ even) and the 3-smooth families $2^g 3^a \mp 1$ in closed form, with second-round readings such as $v_2(a' + 2 - \text{Log}_3(-5))$ — the exponent reading the 2-adic logarithm of a ghost — and every tested seed dies with a finite certificate. The binary world is closed unconditionally by the *entropy deficit*: $\theta_{12}(D) \leq H(2 - \log_2 3) = 0.97907 < 1$ at every band width, so an orbit with letters $\subset \{1, 2\}$ has no phase-diagram refuge anywhere. The weakest nontrivial case of the remaining conjecture is precisely Chang's Carry Independence Conjecture (every $n_0 > 1$ eventually takes a letter $b \geq 3$), which is the parity- $\equiv 1$ extreme of our calibration statement.

4.4 Faithfulness and the blind spot

Is this language adequate? Provably, within its declared scope: the word coding is the Bernstein–Lagarias conjugacy [34, 35] — lossless, a homeomorphism onto the full shift carrying Haar to Bernoulli(2^{-k}); every finite-modulus observable of a bounded window factors exactly through (one residue, the letters); conditional on any word cylinder, residues at all odd places are exactly equidistributed (only the 2-adic place feeds the word); and the empirical hidden-channel test is null (multiplicative invariants of deep survivors match controls; celebrated special numbers — $e^{\pi\sqrt{163}}$'s integer, worst-case Miller–Rabin composites — are Collatz-civilians). The language's one blind spot is the subset $\mathbb{N} \subset \mathbb{Z}_2$ itself, imported through four exact interfaces (clocks, counting, the drift identity,

Baker) — and the blind spot *bites*: a positive seed has $\log_2 x_0$ bits, so after $\approx \frac{2}{3} \log_2 x_0$ steps every letter is forced by the $\times 3$ carry transducer. The whole infinite word has Kolmogorov complexity $\leq \log_2 x_0 + O(1)$ while rigidity demands near-maximal factor complexity: integer-realizable words are a *countable, computable* family (of the 2^{30} binary words of depth 30, sixty-six are realized below 10^6), and survival champions spend 59–76% of their lives in the forced regime, growing like $2^{D/2.2}$ in their survival depth D and behaving exactly like best-of- N random search.

4.5 Convergence of independent frameworks

Tao’s almost-all theory [25], the Chang program [36] (audited piece by piece: its burst-gap calculus *is* the two-clock parse under an explicit dictionary; its census constant verifies; its shadow return-time theorem is refuted and replaced by supply bounds), and this program terminate at the same residual: almost-all versus all; distributional versus pointwise; annealed versus quenched. Three languages, one wall, with explicit translations — the language-invariance expected of a faithful compression of the dynamics.

5 The complete nature of the obstruction

The successive localizations compose into one chain:

Collatz $\leftarrow$ PVT $\longleftrightarrow$ SRCE $\leftarrow$ H23a $\leftarrow$ Quenched Shadow Exclusion $\leftarrow$ Two-Clock Calibration $\leftarrow$.

each arrow an unconditional reduction, terminating in:

The obstruction. Show that no positive integer can feed the round-chain coin even readings with the conspiratorial statistics required for survival, forever. Equivalently: the empirical law of the readings $v_2(m_i - 3^{-\ell_i})$ along a single orbit — a zero-entropy, computable, self-referential stream, provably i.i.d.-fair on Haar fibers and provably lethal on every deterministically analyzable fiber — cannot deviate from its distributional law forever. Every structured deviation channel is metered: counting ($D^* = 1.2463$), bias ceilings (0.449, saturation 0.54), free basin (0.1397), alphabet ($\theta_{12} < 0.9791$), ghost rides ($\lfloor (a-1)/K \rfloor$ with height cost 2^a), supply (2^{k-g} , without replacement), census ($R = 0.0882$ of 0.415), and the Kolmogorov budget ($\log_2 x_0$ bits, spent in the opening). The missing theorem is of distributional-to-pointwise type for one coin.

Three features make this obstruction *complete* as a description: (i) it is *language-invariant* (three independent frameworks reach it under explicit dictionaries); (ii) it is *exhaustive within declared categories* (the finite-shadow language is provably lossless and complete for congruence mechanisms, with the residual — positivity — explicitly isolated, not compressed away); (iii) it is of a *known species*: the same distributional-to-pointwise gap that blocks normality of $\sqrt{2}$ ’s digits, effective Furstenberg $\times 2 \times 3$ rigidity for individual points, and every “true for almost every x , unknown for each x ” statement in metric number theory. The Collatz wall is not an isolated pathology; it is the generic wall of quenched arithmetic dynamics, here presented in its sharpest known finite form.

6 What new mathematics is required

The annealed theory is finished; the congruence category is provably complete; the archimedean imports are exact. What follows is keyed to the surviving attack surfaces, with the literature a reader should enter through.

6.1 Quenched rigidity for self-referential streams (the coin)

The central need: an inverse theorem asserting that a computable zero-entropy stream which persistently deviates from its fiberwise i.i.d. law must exhibit finite algebraic structure (coboundary, ghost-alignment, or periodicity) — structure the certificate machinery then kills. The nearest existing mathematics: Furstenberg’s $\times 2 \times 3$ disjointness and its measure rigidity [8, 9, 10], made *effective* in [11]; Sarnak’s Möbius-disjointness program and the Bourgain–Sarnak–Ziegler criterion [12, 13] as the model for “deterministic sequences cannot correlate with structured ones”; and the Green–Tao–Ziegler inverse theory [14] as the template for “large correlation forces algebraic structure.” None applies as stated: the Collatz cocycle is non-invertible, affine, and self-referential (the stream reads its own output). The new theorem must be an *effective, pointwise* rigidity statement for 2-adic affine transducers — a quenched local limit theorem for one orbit of one machine.

6.2 Digit rigidity and “Baker along orbits”

In the forced regime the letters read binary digits of 3-power multiples of the seed, and the deterministic cascades terminate in readings of 2-adic linear forms in logarithms ($v_2(N - \text{Log}_3(-5))$ and its deeper analogues). The entry points: Stewart’s effective lower bounds for digit counts of 3^n [15], Senge–Straus [16], the p -adic linear-forms theory of Yu [3], Bugeaud’s monograph [4], and — as proof that digit streams *can* be controlled along sparse deterministic sequences — Mauduit–Rivat [17]. The needed extension is quantitative control of v_2 -readings along *orbit-indexed* moving targets: a hybrid of linear forms in logarithms with the renewal structure of the round chain (“Baker along orbits”), for which the periodic case [1, 2, 27, 28] is the solved boundary case.

6.3 Banded S -unit and subspace rigidity (Target 1)

High-complexity banded windows supply many relations $2^B x_{n+p} - 3^p x_n = A_{n,p}$ with $A_{n,p}$ a structured $\{2, 3\}$ -unit sum. The conjecture — too many coherent relations in a multiplicative band force the exponent vectors into finitely many linear patterns, hence repetition or a certificate — is of subspace-theorem type: Evertse–Schlickewei–Schmidt [5] for the count of solutions, Corvaja–Zannier methods [6] for the applications template, Amoroso–Viada [7] for the quantitative state of the art. The obstruction to direct application is that the classical theorems count solutions of *fixed* equations, while the orbit generates a *growing family* of equations with shared structure; what is needed is a rigidity theorem for coherent *families* of S -unit relations.

6.4 Mixed-tower max-plus certificates (Targets 2–3)

The certificate layer — Karp maximum cycle means of $g + \lambda(\phi - \varepsilon)$ over residue traps in $\mathbb{Z}/2^a 3^b$, with the margin $\inf_{\lambda \geq 0} (P_H(\lambda) - \lambda \varepsilon)$ implemented and validated in the candidate tools — needs its deterministic completion: *every positive-realizable coherent trap eventually certifies, except the trivial cycle*. The relevant mathematics is tropical/max-plus spectral theory [24] and ergodic optimization [23]; the missing principle is a projective (lift-coherent) Perron theory for the joint $(2, 3)$ -tower, for which the H23a operator comparison of Campaign I is the prototype.

6.5 The automaton route

The blind-spot theorems reframe the problem as machine behavior: every integer is a finite seed of one fixed carry transducer, and the conjecture asserts the machine always halts (reaches the +1

ghost). The relevant boundary results: Conway’s undecidability for generalized Collatz functions [20, 21] (the family is Turing-complete; any proof must use the specific arithmetic of $3x + 1$), Cobham-type rigidity and automatic-sequence theory [19, 18] (outputs of finite automata cannot be too structured *and* too generic at once — precisely the tension a counterexample must inhabit), and algorithmic randomness [22] for the Kolmogorov-versus-factor-complexity gap. A genuinely new “transducer rigidity” theorem — the output stream of a bounded-carry $\times 3$ transducer from a finite seed cannot be a calibrated biased pseudorandom word forever — would close the conjecture and would be a fundamental statement about multiplication itself.

6.6 Supporting quantitative frame

Tao’s almost-all theorem [25] bounds how large the exceptional set can be; Krasikov–Lagarias density bounds [26] constrain preimage trees; verification stands at 2^{68} [29] with cycle exclusions through $m \leq 91$ [28]; Lagarias’s surveys and sourcebook [31, 32] remain the canonical maps of the terrain; Wirsching’s monograph [33] treats the backward tree; Terras [30] founded the density approach. The contemporary parallel program of Chang [36] is, after our audit, a relabeled fragment of the same structure with the same wall.

7 Endpoint

There is no longer any reason to search for counterexamples: every describable candidate class is excluded by an exact meter, the surviving profile is the output set of one finite machine conditioned on beating a fair coin forever, and a divergent counterexample would in any case be computationally unverifiable. The remaining work is the construction of one theorem — distributional-to-pointwise rigidity for the round-chain coin — toward which the most promising roads are effective $\times 2 \times 3$ -type rigidity, p -adic Baker theory along orbits, S -unit family rigidity, projective max-plus Perron theory, and transducer rigidity. The search for the counterexample and the construction of the proof were the same activity all along: every emptied hiding place above is now a theorem, and the proof, if it comes, will be the completed inventory.

References

- [1] A. Baker, Linear forms in the logarithms of algebraic numbers, *Mathematika* 13 (1966), 204–216.
- [2] G. Rhin, Approximants de Padé et mesures effectives d’irrationalité, in *Séminaire de Théorie des Nombres, Paris 1985–86*, Progr. Math. 71, Birkhäuser, 1987, 155–164.
- [3] K. Yu, p -adic logarithmic forms and group varieties III, *Forum Math.* 19 (2007), 187–280.
- [4] Y. Bugeaud, *Linear Forms in Logarithms and Applications*, IRMA Lect. Math. Theor. Phys. 28, EMS, 2018.
- [5] J.-H. Evertse, H. P. Schlickewei, W. M. Schmidt, Linear equations in variables which lie in a multiplicative group, *Ann. of Math.* 155 (2002), 807–836.
- [6] P. Corvaja, U. Zannier, *Applications of Diophantine Approximation to Integral Points and Transcendence*, Cambridge Tracts in Math. 212, CUP, 2018.

- [7] F. Amoroso, E. Viada, Small points on subvarieties of a torus, *Duke Math. J.* 150 (2009), 407–442.
- [8] H. Furstenberg, Disjointness in ergodic theory, minimal sets, and a problem in Diophantine approximation, *Math. Systems Theory* 1 (1967), 1–49.
- [9] D. Rudolph, $\times 2$ and $\times 3$ invariant measures and entropy, *Ergodic Theory Dynam. Systems* 10 (1990), 395–406.
- [10] B. Host, Nombres normaux, entropie, translations, *Israel J. Math.* 91 (1995), 419–428.
- [11] J. Bourgain, E. Lindenstrauss, P. Michel, A. Venkatesh, Some effective results for $\times a \times b$, *Ergodic Theory Dynam. Systems* 29 (2009), 1705–1722.
- [12] P. Sarnak, Three lectures on the Möbius function, randomness and dynamics, IAS lecture notes, 2011.
- [13] J. Bourgain, P. Sarnak, T. Ziegler, Disjointness of Möbius from horocycle flows, in *From Fourier Analysis and Number Theory to Radon Transforms and Geometry*, Springer, 2013, 67–83.
- [14] B. Green, T. Tao, T. Ziegler, An inverse theorem for the Gowers $U^{s+1}[N]$ -norm, *Ann. of Math.* 176 (2012), 1231–1372.
- [15] C. L. Stewart, On the representation of an integer in two different bases, *J. Reine Angew. Math.* 319 (1980), 63–72.
- [16] H. G. Senge, E. G. Straus, PV-numbers and sets of multiplicity, *Period. Math. Hungar.* 3 (1973), 93–100.
- [17] C. Mauduit, J. Rivat, Sur un problème de Gelfond: la somme des chiffres des nombres premiers, *Ann. of Math.* 171 (2010), 1591–1646.
- [18] B. Adamczewski, Y. Bugeaud, On the complexity of algebraic numbers I. Expansions in integer bases, *Ann. of Math.* 165 (2007), 547–565.
- [19] J.-P. Allouche, J. Shallit, *Automatic Sequences: Theory, Applications, Generalizations*, CUP, 2003.
- [20] J. H. Conway, Unpredictable iterations, in *Proc. 1972 Number Theory Conf.*, Univ. Colorado, Boulder, 1972, 49–52.
- [21] S. A. Kurtz, J. Simon, The undecidability of the generalized Collatz problem, in *TAMC 2007*, LNCS 4484, Springer, 2007, 542–553.
- [22] R. Downey, D. Hirschfeldt, *Algorithmic Randomness and Complexity*, Springer, 2010.
- [23] O. Jenkinson, Ergodic optimization in dynamical systems, *Ergodic Theory Dynam. Systems* 39 (2019), 2593–2618.
- [24] P. Butkovič, *Max-linear Systems: Theory and Algorithms*, Springer, 2010.
- [25] T. Tao, Almost all orbits of the Collatz map attain almost bounded values, *Forum Math. Pi* 10 (2022), e12.

- [26] I. Krasikov, J. C. Lagarias, Bounds for the $3x + 1$ problem using difference inequalities, *Acta Arith.* 109 (2003), 237–258.
- [27] J. L. Simons, B. M. M. de Weger, Theoretical and computational bounds for m -cycles of the $3n + 1$ problem, *Acta Arith.* 117 (2005), 51–70.
- [28] C. Hercher, There are no Collatz m -cycles with $m \leq 91$, *J. Integer Seq.* 26 (2023), Art. 23.3.5.
- [29] D. Barina, Convergence verification of the Collatz problem, *J. Supercomputing* 77 (2021), 2681–2688.
- [30] R. Terras, A stopping time problem on the positive integers, *Acta Arith.* 30 (1976), 241–252.
- [31] J. C. Lagarias, The $3x + 1$ problem and its generalizations, *Amer. Math. Monthly* 92 (1985), 3–23.
- [32] J. C. Lagarias (ed.), *The Ultimate Challenge: The $3x + 1$ Problem*, AMS, 2010.
- [33] G. J. Wirsching, *The Dynamical System Generated by the $3n + 1$ Function*, Lecture Notes in Math. 1681, Springer, 1998.
- [34] D. J. Bernstein, A non-iterative 2-adic statement of the $3N + 1$ conjecture, *Proc. Amer. Math. Soc.* 121 (1994), 405–408.
- [35] D. J. Bernstein, J. C. Lagarias, The $3x + 1$ conjugacy map, *Canad. J. Math.* 48 (1996), 1154–1169.
- [36] E. Y. Chang, Exploring Collatz dynamics with human-LLM collaboration, arXiv:2603.11066 (2026); audited in the ghost-clock companion note of this program.